SELinux Assignment

Task 1: Check the current SELinux status

Step 1: Open the terminal

Switch to the root user:

su -

Step 2: Check the current mode

Run:

getenforce

Possible outputs:

- Enforcing — SELinux actively blocks unauthorized access.
- Permissive — SELinux records policy violations but does not block them.
- Disabled — SELinux is disabled.

Step 3: Check detailed status

sestatus

Look for a line such as:

SELinux status: enabled

Current mode: enforcing

Write down the actual mode shown on your server in your assignment. Don't copy an example as your result.

Task 2: Switch SELinux to Enforcing mode

Run:

setenforce 1

Verify the change:

getenforce

Expected output:

Enforcing

For a non-root user, use sudo setenforce 1.

If the output is Disabled: setenforce cannot enable SELinux. Don't change the configuration or reboot yet; tell me the output first. If the output is already Enforcing, this task is already complete.

Task 3: Simulate an SELinux access denial

We'll create a test webpage under Apache's normal web directory and deliberately assign it the wrong SELinux type.

Step 1: Install and start Apache

Run:

dnf install httpd -y

systemctl enable --now httpd

Verify the service:

systemctl status httpd

Look for active (running).

Step 2: Create a test directory and webpage

mkdir -p /var/www/html/testapp

echo "SELinux Test Page" > /var/www/html/testapp/index.html

chmod 755 /var/www/html/testapp

chmod 644 /var/www/html/testapp/index.html

Check the initial SELinux context:

ls -Zd /var/www/html/testapp

ls -Z /var/www/html/testapp/index.html

The default context may be httpd_sys_content_t or another context, depending on your system.

Step 3: Assign an incorrect SELinux context

Run:

chcon -R -t user_home_t /var/www/html/testapp

Verify the incorrect context:

ls -Zd /var/www/html/testapp

ls -Z /var/www/html/testapp/index.html

The output should now include user_home_t.

Step 4: Try to access the webpage

Run:

curl -i http://localhost/testapp/

If SELinux blocks access, you may see an HTTP 403 Forbidden response. The exact result depends on your Apache configuration and system policy.

Step 5: Check the SELinux audit log

Run:

ausearch -m AVC -ts recent

You can also inspect the audit log directly:

grep 'avc:.*denied' /var/log/audit/audit.log | tail -n 10

Look for an entry containing httpd, denied, and possibly user_home_t.

Important: A 403 response alone does not prove SELinux caused the problem. You need a corresponding SELinux denial in the audit log. If there is no denial, we should check file permissions and Apache logs instead.

Task 4: Correct the SELinux context and resolve the issue

Step 1: Assign the correct context

Run:

chcon -R -t httpd_sys_content_t /var/www/html/testapp

This sets the SELinux type to httpd_sys_content_t, which permits Apache to read static web content under the applicable SELinux policy.

Verify:

ls -Zd /var/www/html/testapp

ls -Z /var/www/html/testapp/index.html

The output should include httpd_sys_content_t.

Step 2: Test the webpage again

Run:

curl -i http://localhost/testapp/

Expected response includes:

HTTP/1.1 200 OK

SELinux Test Page

If you receive 200 OK and the test-page text, the webpage is accessible.

You can also open this URL in the browser on the Linux machine:

http://localhost/testapp/

Step 3: Make the context persistent (recommended)

The chcon command changes the current context, but a future restorecon may reset it. To restore the normal default context for Apache's web directory, run:

restorecon -Rv /var/www/html/testapp

Check the context and test the webpage again. Because this directory is under /var/www/html, the default policy should normally assign an appropriate web-content type.

Task 5: Assignment documentation

Copy this report into your assignment, and fill in the actual outputs from your system.

SELinux Access Control and Troubleshooting

1. Aim

To check the SELinux status, enable Enforcing mode, simulate an access denial by assigning an incorrect SELinux context, and resolve the issue by correcting the context.

2. Checking SELinux Status

Commands used:

getenforce

sestatus

The commands were used to identify the current SELinux mode and obtain detailed status information.

Actual initial mode: **[Write your observed result]**

3. Enabling Enforcing Mode

Command:

setenforce 1

Verification command:

getenforce

Expected result:

Enforcing

Enforcing mode actively blocks actions that violate the SELinux security policy.

4. Simulating an Access Denial

A test directory and HTML file were created in the Apache web directory.

Commands:

mkdir -p /var/www/html/testapp

echo "SELinux Test Page" > /var/www/html/testapp/index.html

The incorrect context was assigned using:

chcon -R -t user_home_t /var/www/html/testapp

The webpage was tested using:

curl -i http://localhost/testapp/

The audit log was checked using:

ausearch -m AVC -ts recent

Observed error or audit entry: **[Paste the actual output, if a denial was recorded]**

5. Correcting the SELinux Context

Command:

chcon -R -t httpd_sys_content_t /var/www/html/testapp

The context was verified using:

ls -Zd /var/www/html/testapp

The webpage was tested again using:

curl -i http://localhost/testapp/

Expected result: HTTP 200 OK with the text SELinux Test Page.

6. How SELinux Contexts Control Access

SELinux assigns security labels to files, directories, and processes. Its security policy determines whether a process such as Apache can access a particular resource. An incorrect file type can prevent Apache from reading a webpage even when normal Linux file permissions allow access. Assigning an appropriate SELinux context resolves the issue when the denial is caused by that context mismatch.

7. Result and Conclusion

SELinux status was checked, and Enforcing mode was verified. An incorrect context was assigned to test web content, and the audit logs were examined for access denials. The context was corrected, and successful webpage access was verified. This practical demonstrated how SELinux provides additional access control beyond traditional Linux file permissions.